

Foto: Dr. Matthias Otte

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EUROfusion

HELMHOLTZ

RESEARCH FOR GRAND CHALLENGES



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IMPURITY TRANSPORT AND RADIATION AT THE STELLARATOR WENDELSTEIN 7-X

Promotionskolloquium

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MOTIVATION AND CONTEXT

NUCLEAR FUSION: THE ENERGY CHALLENGE

- D-T fusion reaction: ${}^2\text{H} + {}^3\text{H} \rightarrow {}^4\text{He}$
(3.5 MeV) + n (14.1 MeV)
- Lawson criterion for net energy gain:

$$n_e T \tau_E \geq \frac{12 f_{\text{tot}}}{\langle \sigma_{\text{DT}} v \rangle f_H^2 E_\alpha - 4 L_Z(T)} T^2$$

- Requires: high temperature ($T \sim$ 10 keV to 20 keV), density, and confinement time
- Magnetic confinement: tokamaks and stellarators

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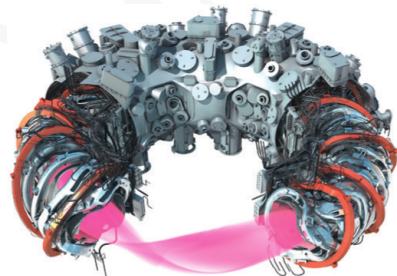
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Triple Product

Key figure of merit for fusion performance combining density, temperature, and energy confinement time

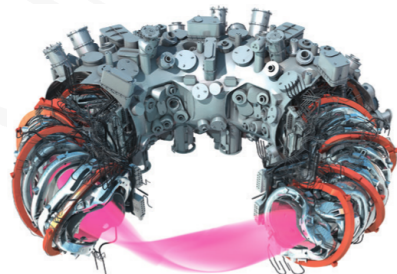
WENDELSTEIN 7-X STELLARATOR

- World's largest stellarator ($R=5.5$ m, $a\sim 0.5$ m)
- 50 superconducting coils, 5-fold symmetry
- Island divertor configuration
- Optimized for reduced neoclassical transport
- Operational phases: OP1.1, OP1.2a, OP1.2b
- Achieved 30 min plasma discharges



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Key Challenge

Managing heat loads on plasma-facing components through controlled radiative cooling

PLASMA RADIATION AND TRANSPORT

Transport Mechanisms:

- Classical: collisional diffusion
- Neoclassical: trapped particles, $\propto 1/\nu$
- Anomalous: turbulent transport (dominant)

Radiation Processes:

- Bremsstrahlung: $P_{\text{Brems}} \propto n_e^2 T^{1/2}$
- Line radiation: atomic transitions
- Impurity effects on plasma performance

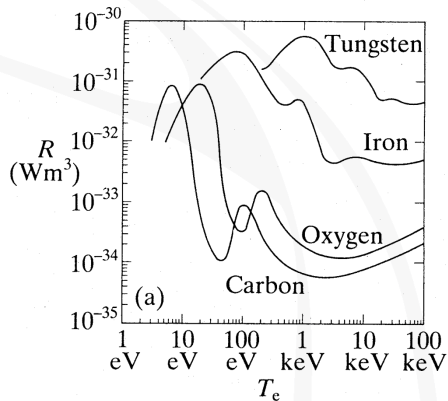
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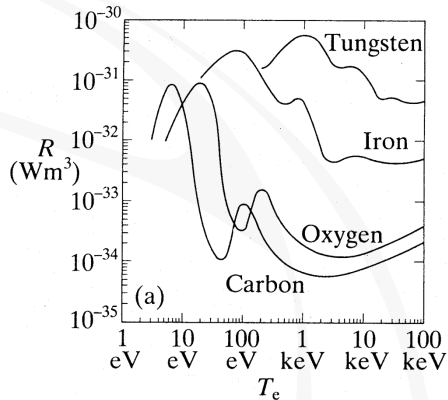
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Deliberate injection of low-Z (He, N₂) or high-Z (Ne, Ar) impurities to enhance

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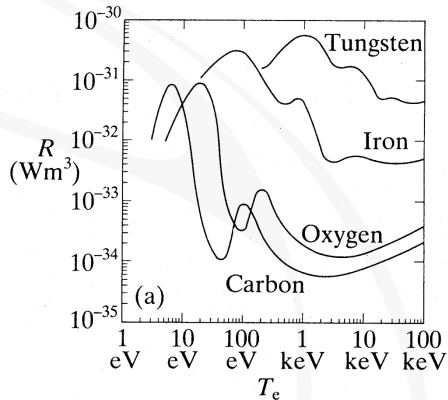
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Radiation Fraction:

$$f_{\text{rad}} = \frac{P_{\text{rad}}}{P_{\text{tot}}}$$



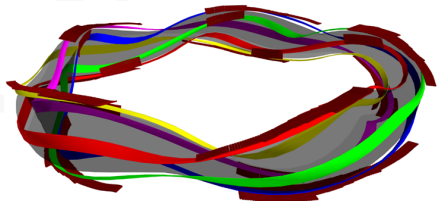
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ISLAND DIVERTOR CONFIGURATION

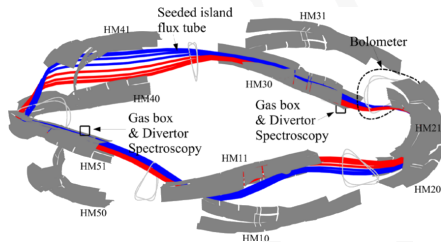
Magnetic Configuration:

- Standard configuration: $\iota = 5/5$
- Five helical island chains
- Ten divertor modules (5 half-modules)
- Field lines connect upstream to downstream targets
- Connection lengths: $\mathcal{O}(100 \text{ m to } 1000 \text{ m})$



Gas Injection Strategy:

- Thermal helium beam valves
- Located behind divertor targets
- Direct seeding into island chains
- Fast piezo valves (ms response)

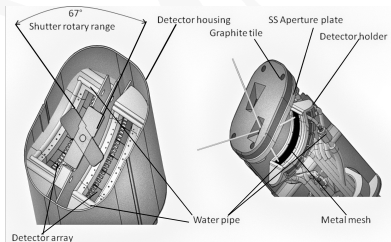


BOLOMETER DIAGNOSTIC SYSTEM

BOLOMETRY AT W7-X

Metal Resistor Bolometers:

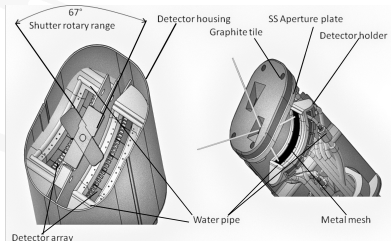
- Thin metal film absorbers (Pt, Au)
- Wheatstone bridge circuit
- Measures total radiation power
- Multicamera system: HBC, VBCI, VBCr
- 128 channels total
- Spatial resolution: ~ 5 cm at magnetic axis



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Measurement Principle:

$$P_M = F_M \cdot \left(\frac{d(\Delta \tilde{U}_M)}{dt} + f_M \Delta \tilde{U}_M \right)$$

Global Radiation Power:

$$P_{\text{rad}} = \frac{V_{P,\text{tor}}}{V_C} \sum_M \frac{P_M V_M}{A_M}$$

BOLOMETER CALIBRATION AND PERFORMANCE

Calibration Methods:

- In-situ laser calibration
- Electrical calibration
- Etendue calculations:
$$K_M = \int_M \widetilde{K}_M dA_M$$
- Volume determination: V_M (LOS cone volume)

Performance Characteristics:

- Time resolution: ~ 1 ms
- Sensitivity: $\sim 10 \text{ kW m}^{-3}$
- Uncertainty: $\sim 10\text{-}15\%$

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Chord Brightness

Line-integrated measurement along each detector's line of sight provides radial profile information

Applications:

- Power balance calculations
- Radiation profile monitoring
- Real-time feedback control
- Tomographic reconstruction

REAL-TIME RADIATION FEED- BACK CONTROL

FEEDBACK SYSTEM DESIGN

System Architecture:

- NI 6321 data acquisition hardware
- LabVIEW control software
- PID controller implementation
- Thermal gas valve actuation
- Minimum latency: 13.6 ms

PID Controller:

$$u(t) = K_p e(t) + K_i \int e(t') dt' + K_d \frac{de(t)}{dt}$$

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Radiation Prediction Proxies:

Subset-based (3-7 channels):

$$P_{\text{pred}}^{(1)} = \frac{V_{\text{P,tor}}}{V_S} \sum_M^S \frac{P_M V_M}{K_M}$$

Single channel (dimensionless):

$$P_{\text{pred}}^{(2)} \propto \Delta U_M$$

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EXPERIMENTAL ACHIEVEMENTS

XP20181010.32 - Benchmark

Discharge:

- Stable helium-seeded detachment
- Radiation fraction: $f_{\text{rad}} > 90\%$
- Peak values up to 100%
- Target heat load reduction: factor of 2
- C^{3+} detachment at $f_{\text{rad}} \sim 50\%$
- Validated 3-channel LOS subset

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Key Result

Demonstrated stable, feedback-controlled radiative cooling with $f_{\text{rad}} \geq 85\%$ without terminal plasma disruption

Comparison with Other Methods:

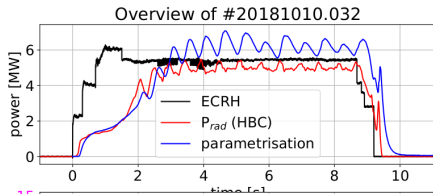
- Electron density feedback
- C-III filterscope feedback
- Bolometer feedback shows intrinsic connection to detachment physics

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FEEDBACK PERFORMANCE ANALYSIS

- **Gas Injection Dynamics:** Moderately scaled gas puffs (medium length and intensity) most effective for reliable edge cooling
- **System Limitations:** Computational and algorithmic constraints introduce non-negligible latencies
- **Optimization Challenges:** Difficult to optimize during commissioning due to limited experimental time

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Critical Finding

The intrinsic connection of P_{rad} and f_{rad} to the detachment process makes bolometer feedback essential for future fusion reactor applications

LINE OF SIGHT SENSITIVITY AND MODELING

IMPURITY SEEDING MODELS

Two-Chamber Model:

$$\dot{N}_w = (N_{w,\text{lim}} - N_w) \tau_{w,p}$$

$$\dot{N}_p = \Gamma_s + N_w \tau_{w,p} - f_{w,p} N_p - N_p \tau_p$$

- Plasma and wall compartments
- Particle exchange rates
- Pumping and recycling

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- Adds SOL compartment
- Better represents edge physics
- Both models equally capable of representing feedback measurements

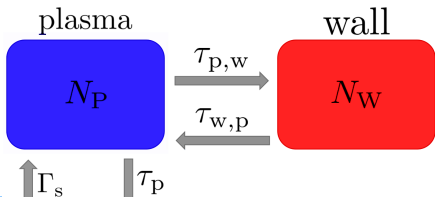
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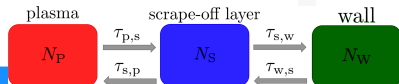
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CHANNEL SELECTION SENSITIVITY

LOS Sensitivity Evaluation:

- Weighted deviation metric
- Correlation analysis
- Mean deviation assessment

Optimal Channel Selection:

- No single "best" set exists
- Robust selection achieves $\geq 85\%$ prediction accuracy
- Agreement between HBC and VBC cameras

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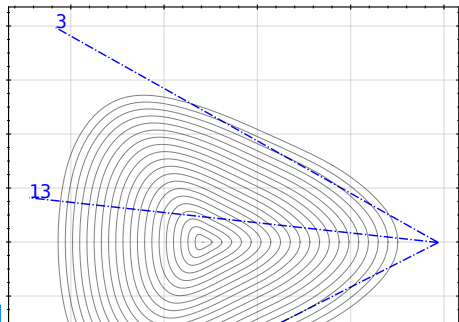
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Key Finding

Detectors viewing separatrix and SOL region most viable for real-time feedback configurations



STRAHL IMPURITY TRANSPORT MODELING

STRAHL Code:

- 1D impurity transport simulation
- Models carbon and oxygen radiation
- Includes ionization, recombination
- Diffusion and convection transport

Key Results:

- Carbon dominates SOL emissivity
- Reduced diffusivity near separatrix
- Inward shift of radiation for $f_{\text{rad}} \rightarrow 1$
- Radiation moves from outside to inside separatrix

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Forward Modeling:

- Calculate chord brightness from STRAHL profiles
- Compare with experimental measurements
- Validate transport assumptions

Discrepancy

Forward calculations show less asymmetry in chord brightness than experiments, suggesting model limitations in capturing 3D effects

TOMOGRAPHIC RECONSTRUCTION

MINIMUM FISHER REGULARIZATION (MFR)

Ill-Posed Inverse Problem:

- 128 line-integrated measurements
- Reconstruct 2D emissivity: ~ 4500 pixels
- Requires regularization

Fisher Information:

$$I_F = \int \frac{1}{g(\vec{r})} \left(\frac{\partial g(\vec{r})}{\partial \vec{r}} \right)^2 d\vec{r}$$

Iterative Solution:

$$\vec{x}^{(n+1)} = \left(\mathbf{T}^T \mathbf{T} + \mu \mathbf{H}^{(n)} \right)^{-1} \mathbf{T}^T \vec{b}$$

Radially Dependent Anisotropy (RDA):

- Tailored weighting: $k_{\text{ani}}(r)$
- $k < 1$: localized structures
- $k > 1$: smooth distributions
- Separate core and edge parameters

MFR Advantages

- Robust to noisy data
- Smooth solutions
- Incorporates a priori knowledge
- Stable convergence

GEOMETRY SENSITIVITY AND BENCHMARKING

Geometry Perturbation Tests:

- Camera position variations
- Detector/aperture segmentation
($N=2,4,8$)
- Triangulation vs rectangular splitting
- Etendue calculation validation

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Key Findings:

- Significant robustness to geometry variations
- $N \geq 8$ segmentation adequate
- Intrinsic bias toward upper SOL/separatrix
- Camera displacement effects non-negligible

GEOMETRY SENSITIVITY AND BENCHMARKING

Phantom Image Benchmarks:

- Simple and complex test profiles
- Quality metrics: χ^2 , ρ_c
- Optimal k_{ani} depends on profile
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Recommended Settings

$k_{\text{ani}} = \{2.0, 0.3\}$ (core, edge)

Grid: 30×150 pixels

Domain: $1.3^2 V_P$

EXPERIMENTAL TOMOGRAPHY RESULTS

Statistical Analysis:

- Forward vs backward P_{rad} correlation
- Core radiation slightly overestimated
- Total power significantly overestimated
- Consistent with phantom benchmarks

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Power Balance Application:

$$P_{\text{bal}} = P_{\text{ECRH}} - P_{\text{rad}} - P_{\text{div}}$$

- $P_{2\text{D}}^{\text{core}}$ improves balance accuracy
- More stable than P_{rad} alone
- Alternative for quasi-steady-state analysis

TOMOGRAPHY QUALITY ASSESSMENT

- **Reconstruction Accuracy:** P_{rad} consistently underestimates total 2D integrated power by amount equal to SOL radiation
- **Quality Metrics:** χ^2 and correlation coefficient ρ_c not always congruent - optimal tomogram may not have $\chi^2 = 1$
- **Anisotropy Optimization:** Ideal k_{ani} set exists for each profile but dimensionality makes finding it impractical
- **Experimental Validation:** Results agree well with phantom benchmarks, supporting algorithm reliability

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Conclusion

MFR with RDA weighting provides robust, reliable 2D radiation reconstructions suitable for physics analysis and power balance calculations, though computational limitations prevent continuous real-time operation

CONCLUSIONS AND OUTLOOK

KEY ACHIEVEMENTS

1. **Real-Time Feedback System:** Successfully implemented, tested and operated first bolometer radiation feedback at W7-X
2. **Line of Sight Optimization:** Systematic sensitivity analysis identified optimal detector configurations
3. **Tomographic Reconstruction:** Developed and benchmarked tailored MFR algorithm with RDA weighting

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3. **Tomographic Reconstruction:** Developed and benchmarked tailored MFR algorithm with RDA weighting
 - Robust to geometry perturbations and noise
 - Successfully applied to experimental data
 - Provides alternative for power balance calculations

SCIENTIFIC INSIGHTS

Detachment Physics:

- C^{3+} detachment visible at $f_{\text{rad}} \sim 50\%$
- Stable detachment achievable at $f_{\text{rad}} > 90\%$
- Radiation shifts inward as $f_{\text{rad}} \rightarrow 1$
- X-point and island radiation concentration

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- No simple scaling law found
- Intrinsic connection to detachment essential
- System latency limits performance

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Diagnostic Performance:

- Multicamera system provides comprehensive coverage
- Geometry uncertainties affect results
- Tomography reveals SOL radiation underestimation

FUTURE OUTLOOK AND RECOMMENDATIONS

System Upgrades:

- Reduce feedback latency through hardware/software optimization
- Implement more complex P_{pred} models
- Per-experiment configurability
- Develop predictive scaling laws

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- Integrate with other SOL models
- Better parametric optimization
- 3D effects in STRAHL modeling

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Tomography Extensions:

- Larger phantom benchmark set
- Additional virtual camera arrays
- Real-time capable algorithms
- Automated parameter optimization

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- Develop predictive scaling laws

Modeling Improvements:

- Refine multi-chamber models with experimental data
- Integrate with other SOL models
- Better parametric optimization
- 3D effects in STRAHL modeling

Tomography Extensions:

- Larger phantom benchmark set
- Additional virtual camera arrays
- Real-time capable algorithms
- Automated parameter optimization

Reactor Relevance:

- Essential for DEMO power plant operation
- Reliable radiative cooling at $f_{\text{rad}} \geq 95\%$
- Machine protection and heat load management
- Steady state operation capability

BROADER IMPACT

Contribution to Fusion Energy Research

This thesis advances the state-of-the-art in plasma diagnostics and control for stellarator fusion devices, providing essential tools and knowledge for achieving reactor-relevant operating scenarios with controlled radiative cooling and stable detachment.

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- First real-time bolometer feedback at W7-X
- Validated diagnostic optimization methods
- Benchmarked tomography algorithm
- Demonstrated stable high- f_{rad} operation

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Technical Contributions:

- First real-time bolometer feedback at W7-X
- Validated diagnostic optimization methods
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- Demonstrated stable high- f_{rad} operation

Scientific Contributions:

- Detachment physics understanding
- Impurity transport insights
- Radiation distribution characterization
- Power balance methodology

SUMMARY

- Developed and operated first real-time bolometer radiation feedback system at W7-X stellarator
- Achieved stable, controlled detachment with radiation fractions $f_{\text{rad}} \geq 85\%$ and peaks up to 100%
- Reduced divertor target heat loads by factor of 2 through controlled impurity seeding
- Validated 3-channel bolometer subset for fast radiation power prediction with $\geq 85\%$ accuracy
- Systematic line of sight sensitivity analysis identified optimal detector configurations for feedback

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- Systematic line of sight sensitivity analysis identified optimal detector configurations for feedback
- STRAHL modeling revealed carbon dominance in SOL radiation and inward shift at high f_{rad}
- Developed and benchmarked Minimum Fisher Regularization tomography with radially dependent anisotropy
- Demonstrated robust 2D radiation reconstruction from experimental data
- Provided alternative power balance methodology using tomographic integrals

COLLABORATIONS

Acknowledgement

This work has been carried out within the framework of the EUROfusion Consortium and has received funding from the Euratom research and training programme 2014-2018 under grant agreement No. 633053. The views and opinions expressed herein do not necessarily reflect those of the European Commission.

Collaborators:



HELMHOLTZ
RESEARCH FOR GRAND CHALLENGES



EUROfusion

Thank You

REFERENCES I

Backup Slides

Additional Details

MEASUREMENT ALGORITHM OVERVIEW

Algorithm Stages:

1. **Initialization** (T1-60 s)
 - Load geometry matrices K_M, V_M
 - Configure channel selection S
 - Initialize analog output
2. **Calibration** (4 s)
 - Offset measurement
 - Two-stage calibration
 - Determine R, τ, κ
3. **Measurement**
 - Continuous voltage integration
 - Moving average filtering
 - Real-time prediction

Real-time Feedback:

- Parallel analog output via NI 6321
- Two prediction channels $P_{\text{pred}}^{(1)}, P_{\text{pred}}^{(2)}$
- FIFO buffer management
- Sample time Δt configurable

Data Storage:

- Local HDD backup
- W7-X archive upload
- Calibration parameters
- Full voltage traces

FOURIER TRANSFORM CORRELATION METRIC

Alternative LOS Sensitivity Evaluation:

Method:

- Uses spectral correlation between P_{rad} and P_{pred}
- Discrete Fast Fourier Transform (DFFT)
- Cross-spectral density analysis
- Coherence-based metric

Mathematical Formulation:

$$\gamma_i = \mathcal{F}(P_{\text{rad}})(\omega)$$

$$\gamma_j = \mathcal{F}(P_{\text{pred}})(\omega)$$

$$\varphi(\omega) = \sqrt{\frac{|g_{i,j}^2|}{|\gamma_i \gamma_j|}}$$

Key Results:

- Similar patterns to correlation metric
- Global maximum at upper inboard X-point
- HBC: best at $0.38r_a$
- VBC: best at $-0.13r_a$
- Spectral focus excludes absolute value contributions

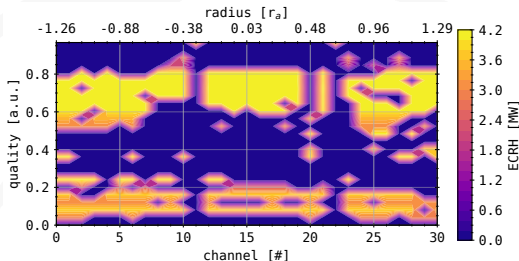
Advantages:

- Frequency domain analysis

Robust to noise

PLASMA PARAMETER SENSITIVITY ANALYSIS

Correlation with Plasma Parameters:



Heating power P_{ECRH} correlation

Key Findings:

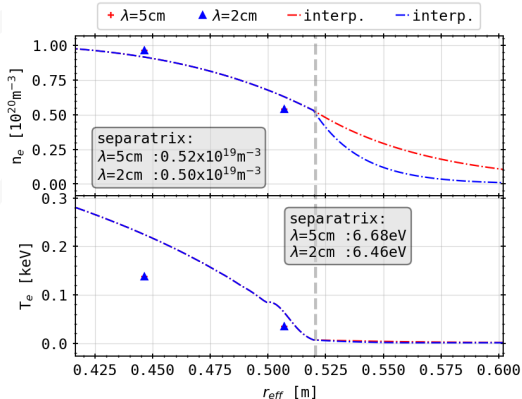
- Higher P_{ECRH} (≥ 3.5 MW) $\rightarrow$ better predictability
- Optimal $f_{\text{rad}} > 0.75$
- Core $T_e \sim 1.3$ keV ideal
- Gap at $\vartheta \approx 0.3-0.5$
- Both cameras in good agreement

Optimal Discharge Conditions:

- Medium-high heating power
- High radiation fraction
- Moderate core temperature
- Consistent across LOS combinations

STRAHL PARAMETER VARIATIONS

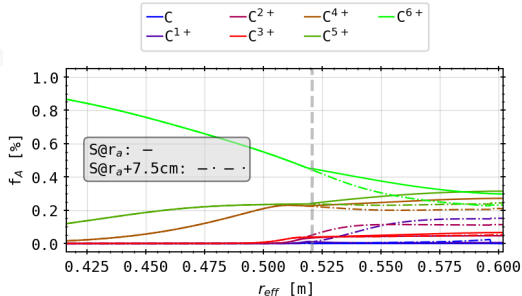
Decay Length Variation:



SOL decay length: $\lambda = 2\text{cm}$ vs 5cm

Impact:

Impurity Source Location:

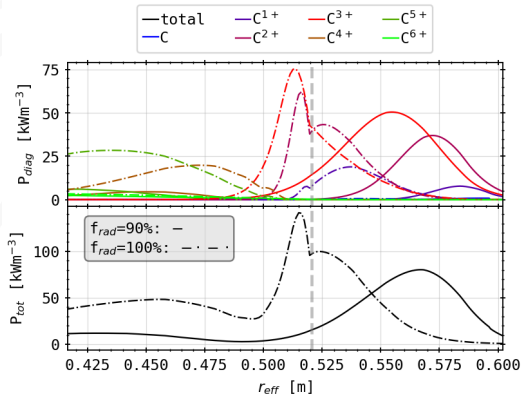


Source at LCFS vs divertor ($r_a+7.5\text{cm}$)

Impact:

- Changes ion populations in SOL
- Similar core emissivity results
- Validates model robustness

STRAHL: RADIATION FRACTION DEPENDENCE



Total radiation for $f_{\text{rad}} = 90\%$ and 100%

Critical Observations:

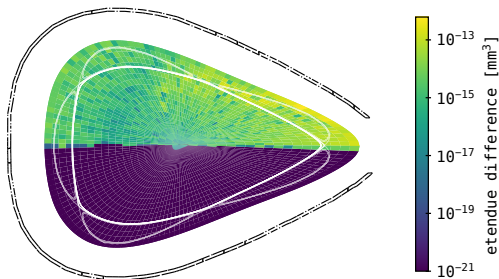
- At $f_{\text{rad}} = 90\%$: peak in SOL
- At $f_{\text{rad}} = 100\%$: peak shifts to LCFS
- C^{3+} and C^{2+} most sensitive
- Dramatic profile changes near separatrix

Implications:

- LOS sensitivity strongly depends on f_{rad}
- Detachment scenarios predictable
- Validates feedback system design
- Critical for controlled detachment

CAMERA GEOMETRY SENSITIVITY

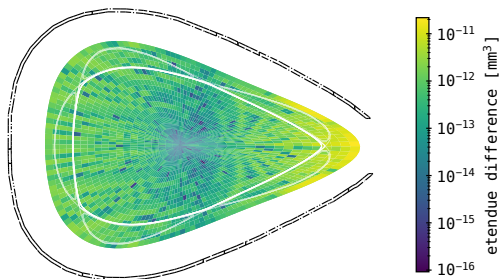
Unilateral Mirroring:



Etendue change: $<10^{-13}$ permm

- Half of HBC array mirrored
- Minimal impact on forward model
- Asymmetry reduced by 0.5-1%

Artificial Symmetric Camera:



Fully symmetric, equidistant array

- Perfectly symmetric geometry
- Similar etendue variations
- Validates as built design

TOMOGRAPHY: RDA VS RGS COMPARISON

Radially Dependent Anisotropy (RDA):

- Standard MFR with k_{ani} profile
- Smooth reconstructions
- Good radial profile matching
- Established method

Relative Gradient Smoothing (RGS):

- Weight: $W_{\text{RGS}} = (1/g_i)^2$
- Can be combined with RDA
- Stronger feature localization
- Higher local intensities

Comparison Results:

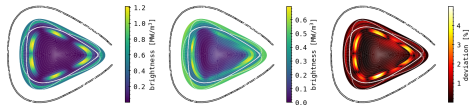
- Similar integrated power values
- RGS: sharper features
- RGS: better peak localization
- RGS: may lose weak structures
- RDA: more robust for general use

Conclusion:

- RDA sufficient for most cases
- RGS useful for specific scenarios
- Combined approach possible
- Choice depends on physics goals

PHANTOM TOMOGRAPHY: EDGE CASES

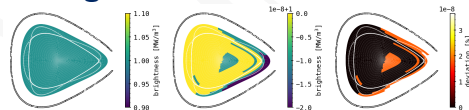
Inverted Anisotropy:



Bright SOL ring + core islands

- More challenging reconstruction
- SOL dominates measurement
- Core features partially obscured

Homogeneous Distribution:



Uniform 1 MW m^{-3} everywhere

- Tests baseline reconstruction
- Validates algorithm behavior
- Confirms no artificial structures

Lessons: Camera sensitivity in SOL can obscure core features; algorithm performs well for uniform distributions; inverted scenarios more challenging than standard core-peaked profiles.

APPENDIX SUMMARY

Measurement Algorithm:

- Three-stage process validated
- Real-time capability confirmed
- Robust calibration procedure
- Full data archiving

LOS Sensitivity Methods:

- Fourier correlation complements time-domain
- Plasma parameter correlations identified
- Optimal discharge conditions determined

STRAHL Validation:

- Parameter variations tested
- Model robustness confirmed
- Critical f_{rad} dependence shown

Tomography Extensions:

- Geometry perturbations minimal
- RDA vs RGS compared
- Edge case phantoms tested
- Algorithm limits characterized